WEEK 3 · LECTURE L1 OF 3 · UNIT I

Data Literacy and Statistical Thinking

How to read an HR number without being fooled.

Part 1 Cold calls on your pod's Round 1 choice

Part 2 Samples and populations, rates and counts, base rates

Part 3 Correlation and causation; common HR data traps

Read Caldwell (2022), data literacy chapters

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PART 1 · COLD CALLS

Three questions on your pod's Round 1 choice

1 · The model

Which maturity model did your pod adopt, and what could it see that the others could not?

2 · The first domain

Which domain did you choose to transform first, and by which criterion?

3 · The test

Which number in your rationale would you now check twice after today's lecture?

THE POINT Any member may be called, not only the Pod Lead. The rationale belongs to the whole pod.

DATA LITERACY, ONE OF THREE

Who is actually in this number?

Population

Everyone you mean to describe. In HR this is rarely what the file contains — leavers drop out, contractors are excluded, new joiners have not been loaded yet.

Sample

The subset you actually have. A sample supports a claim about the population only if you know *how* it was selected.

Self-selection

The dangerous case. Survey respondents chose to answer; app users chose to log in. The people missing are missing *for a reason connected to the thing you are measuring.*

THE CASE THAT MATTERS HERE

480 of 1,000

The GreenLeaf telemetry covers fewer than half the employee records, and it covers the half that **uses the systems**. Any adoption finding drawn from it describes adopters. The non-adopters you are trying to find are the ones the file cannot see.

DATA LITERACY, TWO OF THREE

A count tells you size. A rate tells you risk.

Claim	What it is	Where it misleads
“Store Operations lost the most people”	A count	The largest department almost always loses the most people. The claim may carry no information at all.
“Store Operations has the highest attrition rate”	A rate — leavers over a base	Only as good as the base. Which headcount, measured when, including whom?
“Attrition rose from 31% to 38%”	A rate over time	Check both parts. A falling denominator raises a rate without a single extra person leaving.

Small denominators

One departure from a team of six is a 17% attrition rate. Ranking departments by rate puts small teams at the top and keeps them there.

Report both, always

A rate without its numerator and denominator cannot be checked, and a rate nobody can check is an assertion in decimal form.

PART 2 · DATA LITERACY

An average can hide the whole story

Location tier	Mean digital literacy (1 to 5)	Share scoring 1 or 2	Share scoring 5
Metro (246)	3.88	18%	46%
Tier 1 (225)	3.00	42%	24%
Tier 2 (283)	2.47	60%	10%
Tier 3 (246)	2.23	66%	4%

THE LESSON

The firm-wide mean is 2.88, which describes almost nobody. Two thirds of Tier 3 staff score 1 or 2. Source: Employee_Master, Digital_Literacy_Score by Location_Tier.

L1 · DATA LITERACY

Base rates, and three HR data traps

50%

The base rate trap

A flight-risk flag is 90% accurate. If 10 of 100 people leave, it flags 9 of the 10 leavers, and also 9 of the 90 stayers. So half of everyone flagged was never going to leave.

Survivorship

Exit interviews hear only from leavers; surveys hear only from people who answer. The silent group is often the one you most need.

Simpson's paradox

A gap can reverse when you split the data. A firm-wide pattern may disappear, or flip, inside every department.

Regression to the mean

The worst store this quarter usually improves next quarter anyway. Do not credit the training you happened to run there.

THE HABIT

Before you believe a percentage, ask: out of how many, compared with what, and who is missing from the count?

DATA LITERACY, THREE OF THREE

Four things a correlation can be

Two HR variables move together. Before you write a recommendation, rule out the three explanations that are not the one you want.

A causes B

The claim you want. It needs more than co-movement: a mechanism, and ideally variation you did not choose.

B causes A

Reverse causation. Training completion correlates with engagement — and engaged people volunteer for training.

C causes both

A confounder. Grade band drives device access *and* literacy *and* app usage. Three correlations, one cause.

Selection made it

The correlation lives in how the file was built, not in the world. See the previous slide.

Why this is not academic fussiness: an intervention is a causal bet. If you spend on training because training correlates with engagement, and the true story is that engaged people seek out training, you will spend the budget and watch nothing move.

PART 3 · THREE MISTAKES THAT COST MARKS

Percentage points, full scale, and the right denominator

Points, not percent

A rate rising from 20% to 30% is a rise of 10 percentage points, but 50 percent. Say which one you mean.

Sample or full scale

Employee_Master holds 1,000 of 8,000 people. A Board wants the full-scale figure: multiply the sample count by 8.

The right base

Adoption_Telemetry has 480 rows, not 1,000. Divide by the people actually in the table.

BEFORE YOU SUBMIT

Read every number aloud with its unit and its base: "34.6 percentage points, among 300 partners."

Going deeper: the theory underneath data literacy

The next slides set out base-rate neglect, Simpson's paradox, the potential outcomes view of causation, and the Bradford Hill criteria: what each claims, how it works, where it breaks, and what it means at GreenLeaf.

THEORY IN DEPTH · KAHNEMAN AND TVERSKY (1973)

We ignore base rates

THE CORE IDEA

People judge probability by how well a case fits a story, and neglect how common the outcome is.

HOW IT WORKS

A vivid profile overrides the base rate. The fix is to start from the base rate, then adjust.

LIMITS AND CRITIQUE

Base rates are used better when problems are framed as natural frequencies (Gigerenzer).

AT GREENLEAF

A flight-risk flag must be read against GreenLeaf's actual attrition rate, or most flagged people will be false alarms.

THEORY IN DEPTH · BICKEL, HAMMEL AND O'CONNELL (1975)

Simpson's paradox: the aggregate can lie

THE CORE IDEA

UC Berkeley graduate admissions appeared to favor men overall.

HOW IT WORKS

Department by department, women were admitted at similar or slightly higher rates. Women had applied to more competitive departments.

LIMITS AND CRITIQUE

The right level of analysis depends on the causal question, not on the data alone.

AT GREENLEAF

A firm-wide gender gap in offers must be checked within each role and tier before drawing a conclusion.

THEORY IN DEPTH · RUBIN (1974); HOLLAND (1986)

Causation means comparing with the counterfactual

THE CORE IDEA

The effect of training on a person is the difference between their outcome with and without it.

HOW IT WORKS

We see only one outcome, so we compare groups. Random assignment makes the groups comparable.

LIMITS AND CRITIQUE

Without randomization, hidden differences between groups can create a false effect.

AT GREENLEAF

Staff who chose training may differ from those who did not. Compare like with like before crediting the course.

THEORY IN DEPTH · BRADFORD HILL (1965)

Nine tests for judging a causal claim

THE CORE IDEA

When experiments are impossible, weigh several kinds of evidence for causation.

HOW IT WORKS

Strength, consistency, specificity, temporality, dose-response, plausibility, coherence, experiment and analogy.

LIMITS AND CRITIQUE

Only temporality is strictly necessary; the rest are guides, not a checklist to sum up.

AT GREENLEAF

Does a higher engagement score come before lower attrition, and do more engaged teams leave less? Test temporality first.

BRIDGE TO L2

Three things to carry into business intelligence

1

Every rate has a numerator and a denominator. Check both.

2

Means hide spreads. Split by segment before you conclude.

3

Correlation is a clue, not a cause. Rule out the rivals.

NEXT

If numbers can mislead, who decides how they are defined? That is the job of a metric layer.